

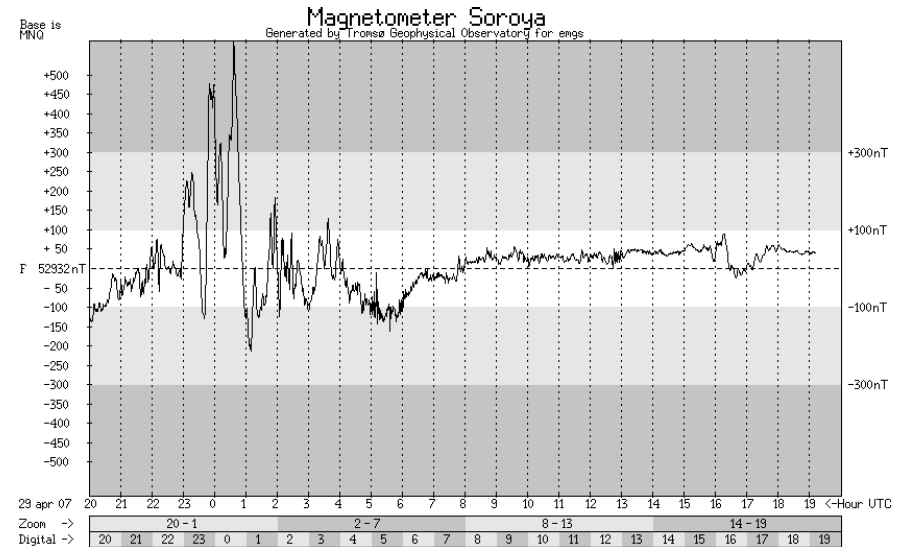
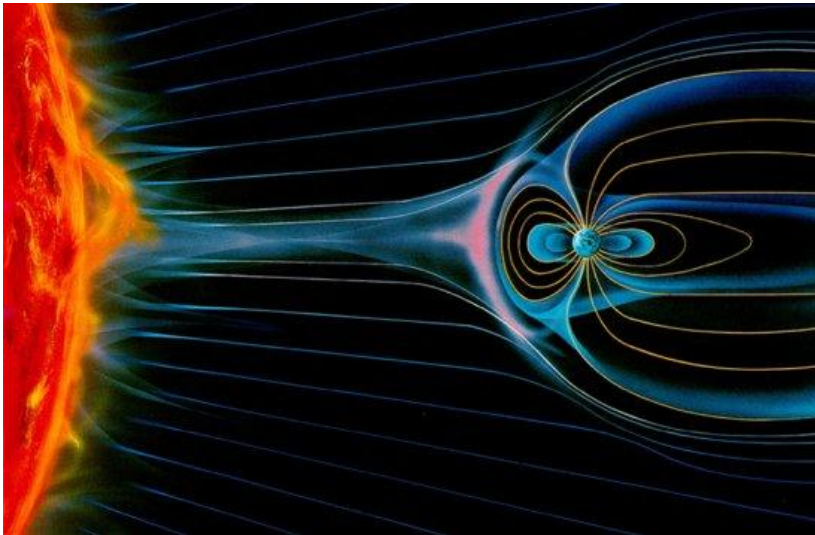
Magnetotellurics

MT inversion

Ketil Hokstad, March 2016

Magnetotellurics (MT)

Tikhonov (1950)
Cagniard (1953)



- Passive exploration method
- Naturally occurring geomagnetic variations is the power source
- Measurement of the scattered MT field on the surface, at different frequencies, allows for resistivity vs depth imaging

Applications of MT

- Salt imaging
- Basalt imaging
- Geothermal exploration
- Ore exploration (mining)
- Crustal studies
- Mantle studies

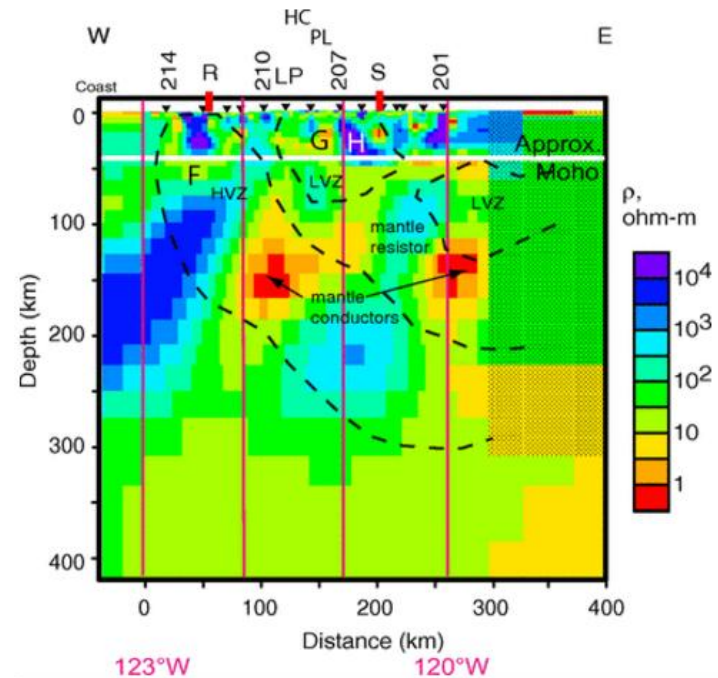
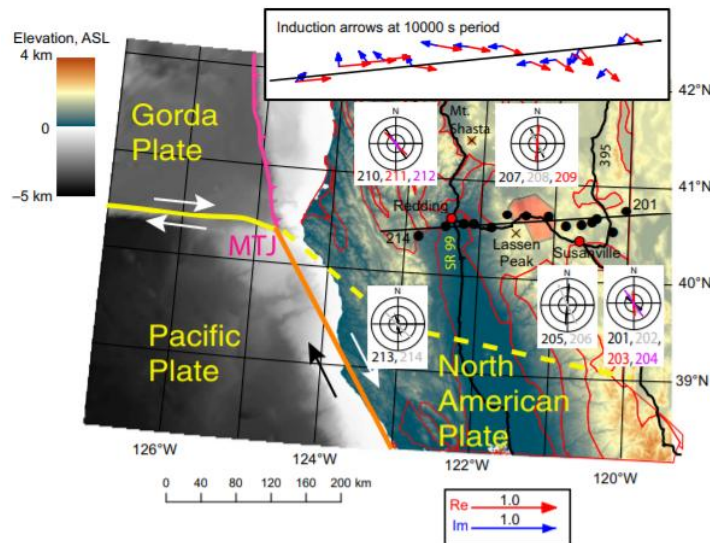


Figure 6.

Best-fit unconstrained model for the magnetotelluric profile. Resistivity section is plotted with no vertical exaggeration. Heavy white line denotes approximate position of Mohorovičić discontinuity from receiver function study (Frassetto et al., 2011). High-velocity zones (HVZs) and low-velocity zones (LVZs) from seismic tomography section in Figure 7 are shown. Area F denotes the conductive region at 50 km depth beneath Redding (R). Areas G and H correspond to conductive and resistive regions of the crust, respectively. Dotted pattern on eastern edge of mantle shows area of constrained resistivities for inversion. Mantle conductors and resistor are mentioned in text. Longitude lines are shown on the section for reference. See caption of Figure 3 for explanation of symbols.

Stephen K. Park, and Linda C. Ostos
Geosphere 2013;9:382-393

CSEM and MT acquisition April 2007



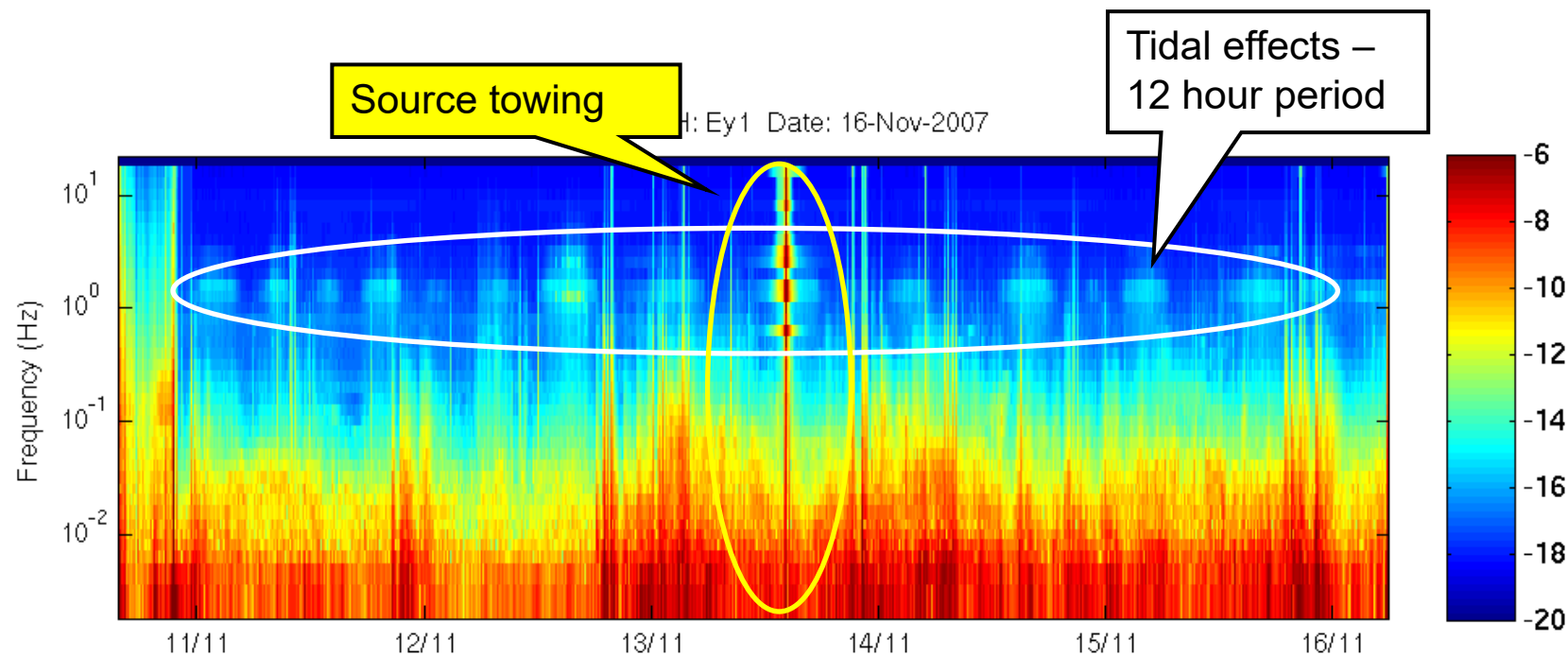
EM acquisition with emgs, Nordkapp Basin April-May 2007



Photo: K.Hokstad

- CSEM: Purpose designed high-frequency source signal
- MT: 48 hours nominal listening time

MT spectrogram from on-board QC



$$\text{Apparent resistivity: } \rho_{ij}^A = \frac{1}{\omega\mu_0} |Z_{ij}|^2$$

Impedance tensor

3D medium:

$$E_I = \sum_J Z_{IJ} H_J, \quad (I, J = 1, 2)$$

2D medium:

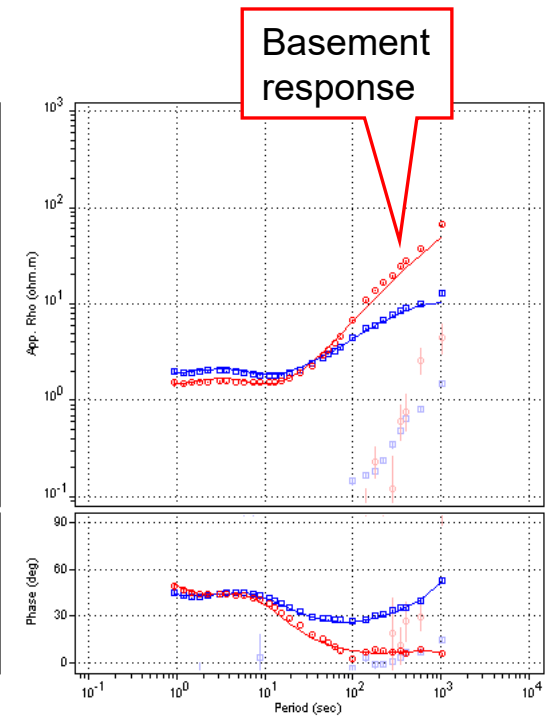
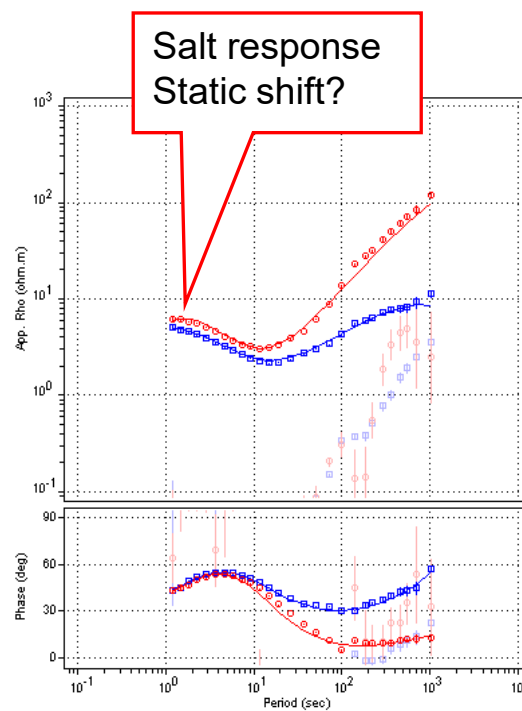
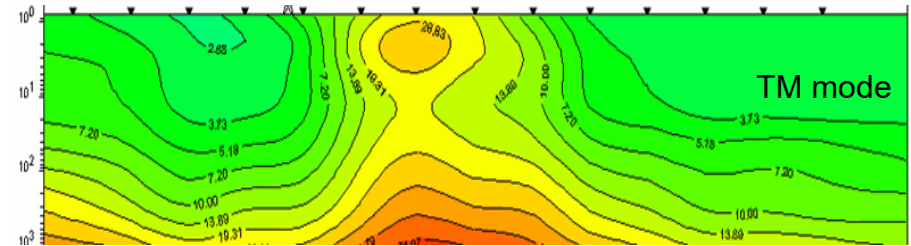
$$Z_{xy} = \frac{E_x}{H_y}$$

$$Z_{yx} = \frac{E_y}{H_x}$$

1D medium:

$$Z_{xy} = \frac{E_x}{H_y}$$

$$Z_{yx} = -Z_{xy}$$



Data example from Nordkapp Basin
Data processing by Geosystem

TM mode = currents $\perp$ strike

TE mode = currents $\parallel$ strike

Skin depth, apparent resistivity

Skin depth:

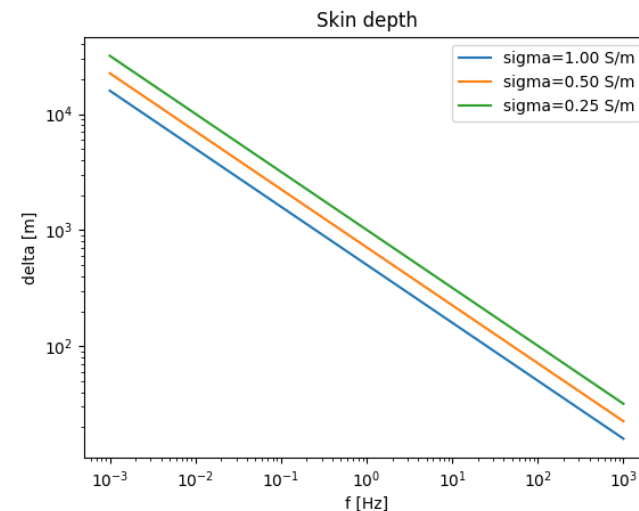
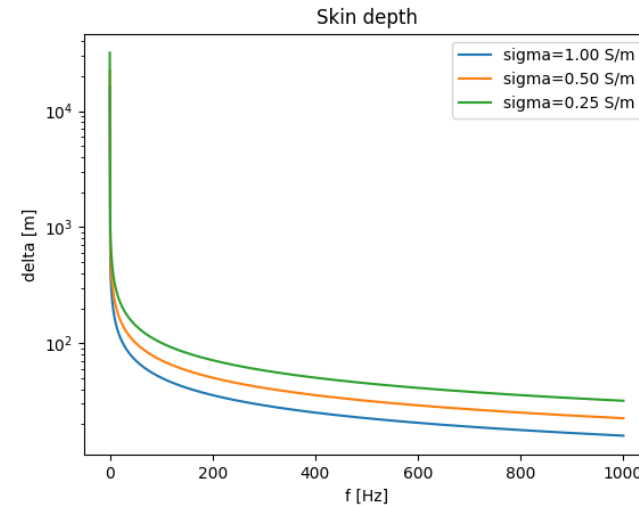
$$E \propto e^{-(1+i)z/\delta}$$

$$\delta = \sqrt{\frac{1}{\pi\mu_0\sigma f}} = \sqrt{\frac{T}{\pi\mu_0\sigma}}$$

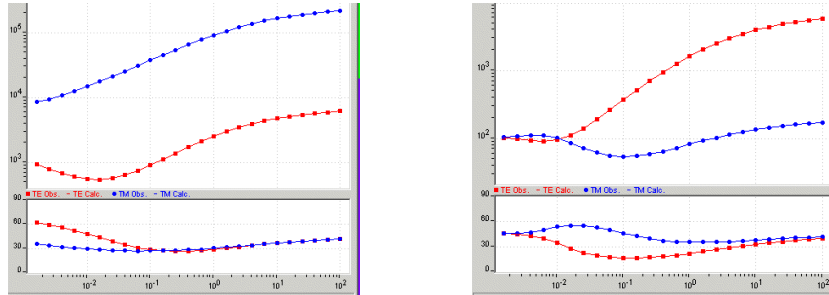
Apparent resistivity:

$$\rho_{ij}^A = \frac{1}{\omega\mu_0} |Z_{ij}|^2$$

$$\phi_{ij}^A = \tan^{-1} \frac{\text{Im}(Z_{ij})}{\text{Re}(Z_{ij})}$$



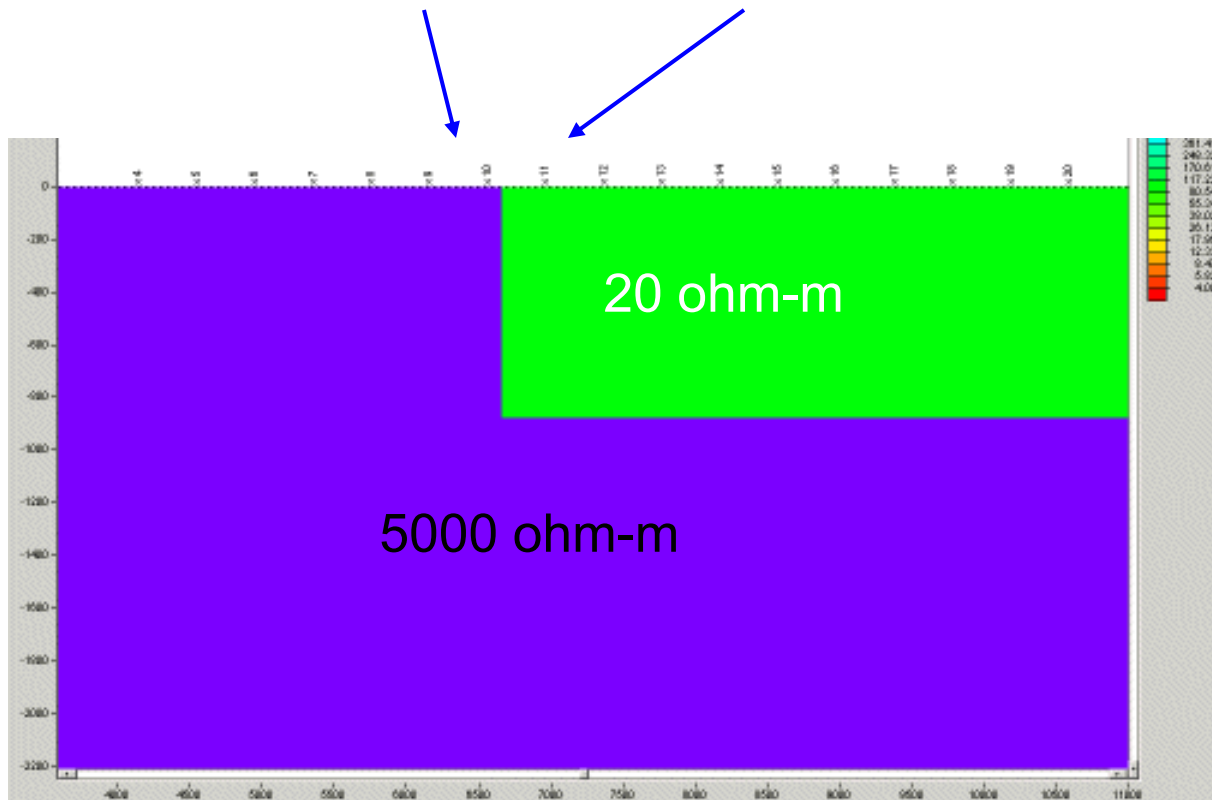
MT interpretation - boundary



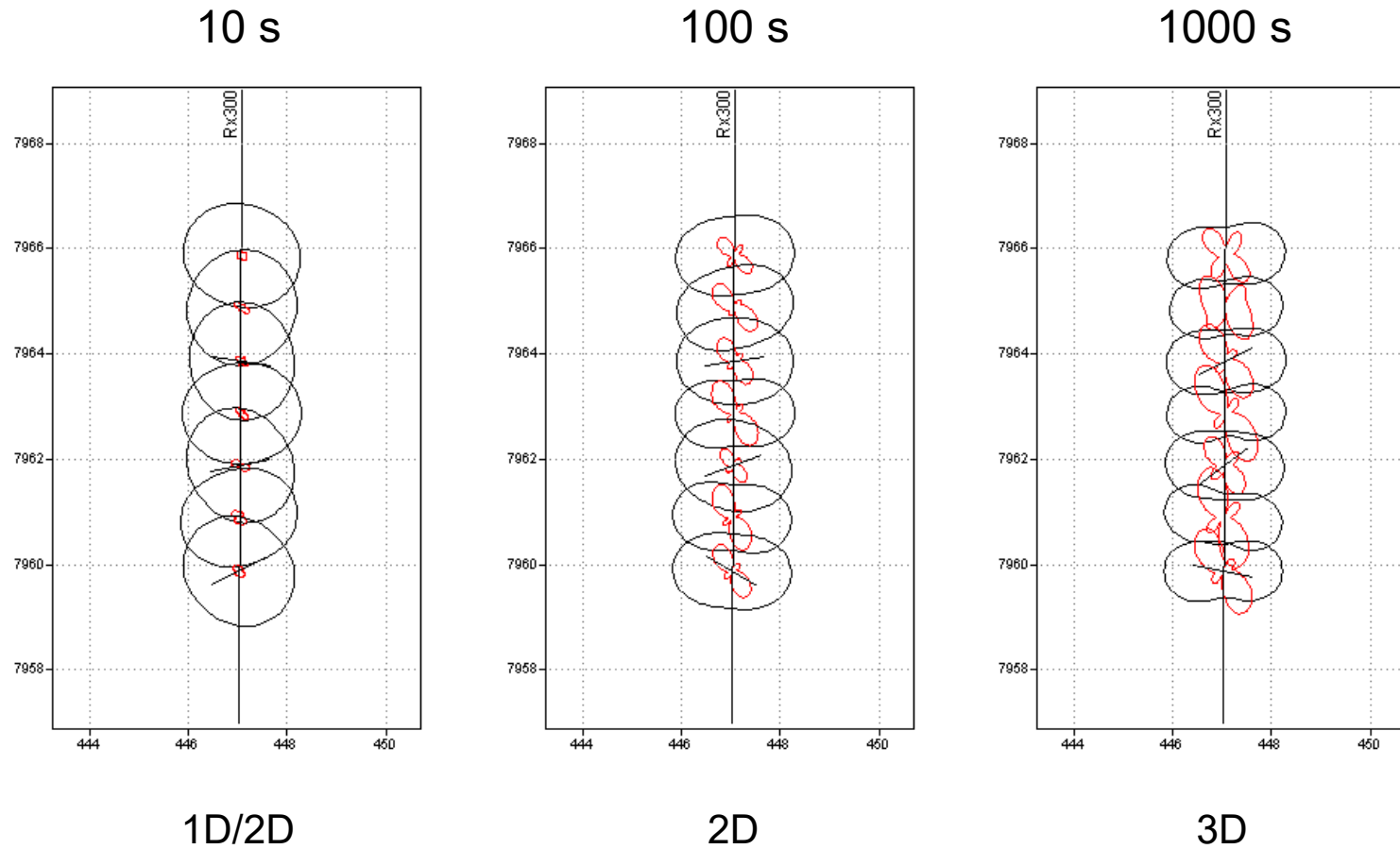
Current continuity across boundary

$$\sigma_1 E_1 = \sigma_2 E_2$$

For the normal component



MT interpretation - dimensionality



Data from Nordkapp Basin

MT inversion (Mütschard et al. 2016)

3D forward problem:

$$\nabla \times \nabla \times \mathbf{E} + i\omega\mu_0\sigma\mathbf{E} = -i\omega\mu_0\mathbf{J}_s$$

1D background field (analytical solution):

$$\nabla \times \nabla \times \mathbf{E}^0 + i\omega\mu_0\sigma_0\mathbf{E}^0 = -i\omega\mu_0\mathbf{J}_s$$

Scattered field:

$$\mathbf{E}' = \mathbf{E} - \mathbf{E}^0$$

3D forward problem for the scattered field:

$$\nabla \times \nabla \times \mathbf{E}' + i\omega\mu_0\sigma\mathbf{E}' = -i\omega\mu_0(\sigma - \sigma_0)\mathbf{E}^0$$

Magnetic field (Faraday):

$$\mathbf{B} = \mu_0\mathbf{H} = -\frac{1}{i\omega}\nabla \times \mathbf{E} = -\frac{1}{i\omega}\nabla \times (\mathbf{E}^0 + \mathbf{E}')$$

Maxwell:

$$\nabla \times \mathbf{E} = -\mu_0\partial_t\mathbf{H}$$

$$\nabla \times \mathbf{H} = \mathbf{j}^E$$

Ohm's law:

$$\mathbf{j}^E = \sigma\mathbf{E}$$

Recap: Matrix-form of 1st and 2nd derivatives (1D)

$$\partial_z^+ u_k \simeq \frac{u_{k+1} - u_k}{\Delta z}$$

$$\partial_z^2 u_k \simeq \frac{u_{k+1} - 2u_k + u_{k-1}}{\Delta z^2}$$

$$\partial_z^+ \mathbf{u} \simeq -\frac{1}{\Delta z} \mathbf{D} \mathbf{u}$$

$$\partial_z^2 \mathbf{u} \simeq -\frac{1}{\Delta z^2} \mathbf{D}^T \mathbf{D} \mathbf{u}$$

$$\mathbf{D} = \begin{bmatrix} 1 & -1 & & & \\ & 1 & -1 & & \\ & & 1 & -1 & \\ & & & 1 & -1 \\ & & & & 1 \end{bmatrix}$$

$$\mathbf{D}^T \mathbf{D} = \begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & -1 & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{bmatrix}$$

in one spatial dimension

Recap: Least-squares and minimum norm

Model update if $\mathbf{J}^H \mathbf{J}$ or $\mathbf{J} \mathbf{J}^H$ can be inverted

$$\Delta \mathbf{m} = (\mathbf{J}^H \mathbf{J})^{-1} \mathbf{J}^H \Delta \mathbf{d} \quad (\text{least squares})$$

$$\Delta \mathbf{m} = \mathbf{J}^H (\mathbf{J} \mathbf{J}^H)^{-1} \Delta \mathbf{d} \quad (\text{minimum norm})$$

Moore-Penrose
pseudo inverse

$$\mathbf{m}^{(n+1)} = \mathbf{m}^{(n)} + \Delta \mathbf{m}$$

The linear inverse problem revisited

$$\mathcal{F}(\mathbf{m}) = \mathbf{A} \mathbf{m}$$

$$\frac{\partial \mathcal{F}}{\partial \mathbf{m}} = \mathbf{J} = \mathbf{A}$$

Forward problem, matrix-vector form :

$$\mathbf{A}\mathbf{e} = \mathbf{b}$$

$$\mathbf{A} = \text{coefficient matrix (13 entires per row)}$$

$$\mathbf{e} = [E'_i(\mathbf{x})] \text{ (scattered field)}$$

$$\mathbf{b} = [-i\omega\mu_0(\sigma - \sigma_0)\mathbf{E}_0] \text{ (source; TE and TM)}$$

Objective function:

$$\phi = \frac{1}{\lambda} \left[(\mathbf{d} - \mathcal{F}(\mathbf{m}))^T \mathbf{C}_d^{-1} (\mathbf{d} - \mathcal{F}(\mathbf{m})) \right] + (\mathbf{m} - \mathbf{m}_0)^T \mathbf{C}_m^{-1} (\mathbf{m} - \mathbf{m}_0)$$

$$\mathbf{d} = [\mathbf{E}^{obs}(\mathbf{x}_r)] \text{ or } [\mathbf{Z}^{obs}(\mathbf{x}_r)]$$

$$\mathbf{m} = [\sigma(\mathbf{x})]$$

Minimum-norm model update:

$$\mathbf{m}_{k+1} - \mathbf{m}_k = \mathbf{C}_m \mathbf{J}_k^T (\lambda \mathbf{C}_d + \mathbf{J}_k \mathbf{C}_m \mathbf{J}_k^T)^{-1} (\mathbf{d} - \mathcal{F}(\mathbf{m}_k) + \mathbf{J}_k(\mathbf{m}_k - \mathbf{m}_0))$$

"Data space" Hessian:

$$\begin{aligned} \mathbf{H} &\simeq (\lambda \mathbf{C}_d + \mathbf{J}_k \mathbf{C}_m \mathbf{J}_k^T) \\ \dim(\mathbf{H}) &= N_d \times N_d \ll N_m \times N_m \end{aligned}$$

Minimum-norm has computational advantages

Jacobi matrix:

$$\begin{aligned} \mathbf{J} &= \frac{\partial \mathcal{F}}{\partial \mathbf{m}} \\ \mathbf{J}^T &= \left(-\frac{\partial \mathbf{A}}{\partial \mathbf{m}} \mathbf{e} + \frac{\partial \mathbf{b}}{\partial \mathbf{m}} \right)^T (\mathbf{A}^{-1})^T \left(\frac{\partial \psi}{\partial \mathbf{e}} \right)^T \end{aligned}$$

Scattered-field source
depends on model $\Rightarrow \frac{\partial \mathbf{b}}{\partial \mathbf{m}} \neq 0$

Full-waveform inversion:

$$\mathbf{J}^H = - \left(\mathbf{B}^{-1} \frac{\partial \mathbf{B}}{\partial \mathbf{m}} \mathbf{u} \right)^H$$

Function ψ maps $[\mathbf{E}, \mathbf{H}]$ to data $\mathbf{d}$

Some more details of the differentiations

$$\mathbf{A}\mathbf{e} = \mathbf{b}$$

$$\mathbf{d} = \boldsymbol{\psi}(\mathbf{e}(\mathbf{m})) + \mathbf{n}$$

$$\frac{\partial \mathbf{A}}{\partial \mathbf{m}} \mathbf{e} + \mathbf{A} \frac{\partial \mathbf{e}}{\partial \mathbf{m}} = \frac{\partial \mathbf{b}}{\partial \mathbf{m}}$$

$$\mathbf{J} = \frac{\partial \boldsymbol{\psi}}{\partial \mathbf{m}} = \frac{\partial \boldsymbol{\psi}}{\partial \mathbf{e}} \frac{\partial \mathbf{e}}{\partial \mathbf{m}}$$

$$\frac{\partial \mathbf{e}}{\partial \mathbf{m}} = \mathbf{A}^{-1} \left(-\frac{\partial \mathbf{A}}{\partial \mathbf{m}} \mathbf{e} + \frac{\partial \mathbf{b}}{\partial \mathbf{m}} \right)$$

$$\mathbf{J}^T = \left[\frac{\partial \boldsymbol{\psi}}{\partial \mathbf{e}} \mathbf{A}^{-1} \left(-\frac{\partial \mathbf{A}}{\partial \mathbf{m}} \mathbf{e} + \frac{\partial \mathbf{b}}{\partial \mathbf{m}} \right) \right]^T$$

3D impedance:

$$Z_{xy} = \frac{E_x^{(2)} H_x^{(1)} - E_x^{(1)} H_x^{(2)}}{H_x^{(1)} H_y^{(2)} - H_x^{(2)} H_y^{(1)}}$$

Nasty equation;
don't read it



Derivative of ψ for impedance:

$$\begin{aligned} \frac{\partial Z_{xy}}{\partial e_k} = & \left[\frac{\partial E_x^{(1)}}{\partial e_k} H_x^{(2)} (H_x^{(2)} H_y^{(1)} - H_x^{(1)} H_y^{(2)}) + \frac{\partial H_x^{(1)}}{\partial e_k} H_x^{(2)} (E_x^{(1)} H_y^{(2)} - E_x^{(2)} H_y^{(1)}) \right. \\ & + \frac{\partial H_y^{(1)}}{\partial e_k} H_x^{(2)} (E_x^{(2)} H_x^{(1)} - E_x^{(1)} H_x^{(2)}) + \frac{\partial E_x^{(2)}}{\partial e_k} H_x^{(1)} (H_x^{(1)} H_y^{(2)} - H_x^{(2)} H_y^{(1)}) \\ & \left. + \frac{\partial H_x^{(2)}}{\partial e_k} H_x^{(1)} (E_x^{(2)} H_y^{(1)} - E_x^{(1)} H_y^{(2)}) + \frac{\partial H_y^{(2)}}{\partial e_k} H_x^{(1)} (E_x^{(1)} H_x^{(2)} - E_x^{(2)} H_x^{(1)}) \right] / (H_x^{(1)} H_y^{(2)} - H_x^{(2)} H_y^{(1)})^2 \end{aligned}$$

Derivative of ψ for fields:

$$\frac{\partial \psi}{\partial \mathbf{e}} = \frac{\partial \mathbf{E}_x}{\partial e_k} = \frac{\partial \mathbf{E}_x^{(1)}}{\partial e_k} + \frac{\partial \mathbf{E}_x^{(2)}}{\partial e_k}$$

However, to invert fields, we need to ' know the source (next)!

MT source estimation (plane wave)

Let $e_q^{(j)}$ be due to a unit source, then

$$E_q(\mathbf{x}, \omega) = \sum_{j=1}^2 \alpha^{(j)}(\omega) e_q^{(j)}(\mathbf{x}, \omega)$$

Kirchhoff integrals for $e_m^{(j)}$ and $E_m^{(j)}$

$$e_m^{(j)}(\mathbf{r}, \omega) = -2 \int ds n_i \epsilon_{ijk} G_{mj}^E(\mathbf{r} | \mathbf{r}_s) h_k^{(j)}(\omega)$$

$$h_k^{(j)} = \text{unit magnetic surface source}$$

$$j = 1, 2 \quad (= TM, TE)$$

Linear inverse problem (over determined)

$$\mathbf{p} = \mathbf{L}\mathbf{q} + \mathbf{n}$$

Einstein sum convention:

$$\vec{a} \cdot \vec{b} = \sum a_i b_i = a_i b_i$$

$$[\vec{a} \times \vec{b}]_i = \epsilon_{ijk} a_j b_k$$

ϵ_{ijk} = Levi-Civita tensor

$$\epsilon_{ijk} = -\epsilon_{jik} \text{ (antisymmetric)}$$

$$\epsilon_{ijk} = \epsilon_{kij} \text{ (cyclic)}$$

$$\epsilon_{ijk} \epsilon_{ilm} = \delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}$$

$$\mathbf{p} = \begin{bmatrix} E_x(\mathbf{r}_1, \omega) \\ E_y(\mathbf{r}_1, \omega) \\ \dots \\ E_x(\mathbf{r}_n, \omega) \\ E_y(\mathbf{r}_n, \omega) \end{bmatrix} \quad \mathbf{L} = \begin{bmatrix} e_x^{(1)}(\mathbf{r}_1, \omega) & e_x^{(2)}(\mathbf{r}_1, \omega) \\ e_y^{(1)}(\mathbf{r}_1, \omega) & e_y^{(2)}(\mathbf{r}_1, \omega) \\ \dots & \dots \\ e_x^{(1)}(\mathbf{r}_n, \omega) & e_x^{(2)}(\mathbf{r}_n, \omega) \\ e_y^{(1)}(\mathbf{r}_n, \omega) & e_y^{(2)}(\mathbf{r}_n, \omega) \end{bmatrix} \quad \mathbf{q} = \begin{bmatrix} \alpha^{(1)}(\omega) \\ \alpha^{(2)}(\omega) \end{bmatrix}$$

Pseudo inverse (least squares):

$$\begin{aligned} \mathbf{q} &= [(\mathbf{WL})^T (\mathbf{WL}) + \lambda \mathbf{S}]^{-1} (\mathbf{WL})^T (\mathbf{Wp}) \\ \mathbf{S} &= \text{diag}(\mathbf{L}^T \mathbf{L}) \end{aligned}$$

Magnetic boundary condition for the forward problem:

$$H_i(\mathbf{x}_s, \omega) = \sum_{j=1}^2 \alpha^{(j)}(\omega) h_i^{(j)}(\mathbf{x}, \omega)$$

MT impedance tensor vs fields

Impedance tensor $\mathbf{Z}$:

$$Z_{xy} = \frac{E_x}{H_y}$$

Advantages:

- Deconvolution for unknown spectrum
- Impedance elements can be interpreted directly

Disadvantages:

- Makes the inverse problem more non-linear ($\mathbf{Z}$ is not a physical field)

Fields $\mathbf{E}$ and $\mathbf{H}$:

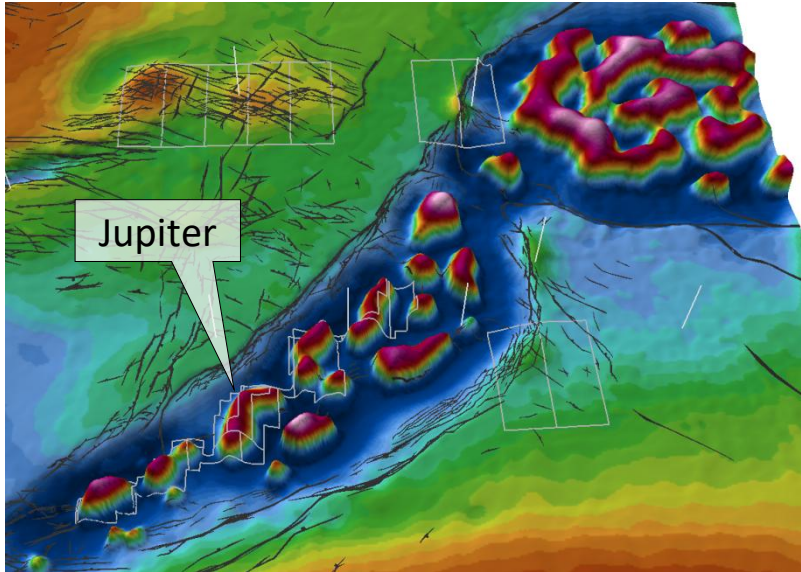
Advantages:

- Maxwell equations are for $\mathbf{E}$ and $\mathbf{H}$
- Easier formulation of the inverse problem

Disadvantages:

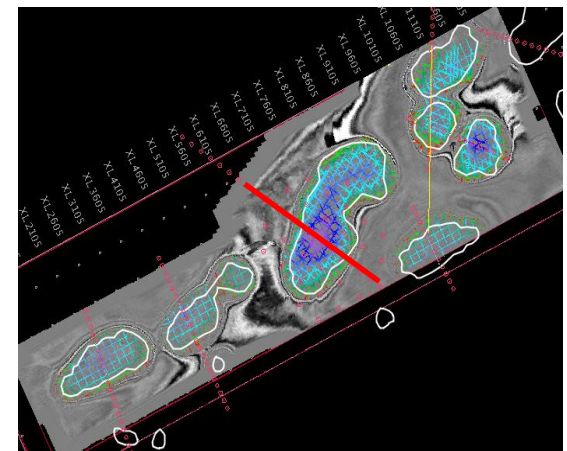
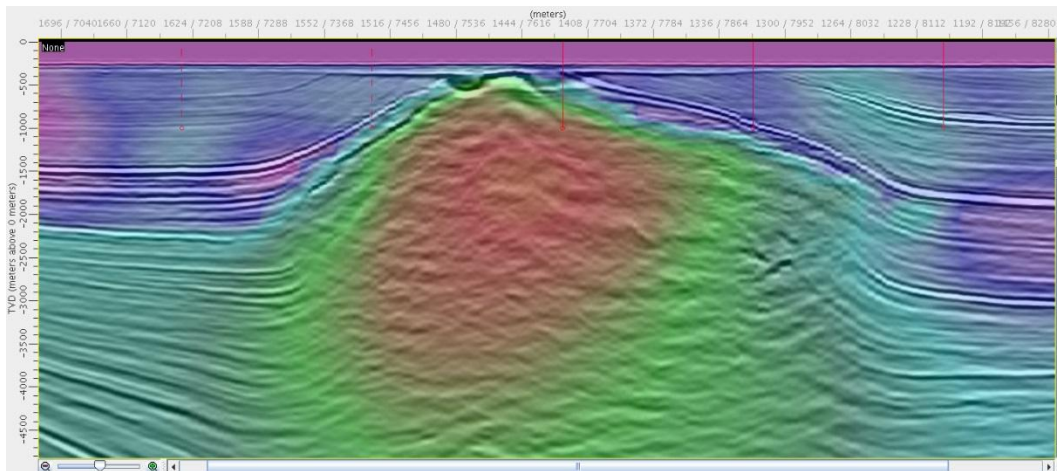
- Need to estimate the MT source spectrum

Nordkapp Basin

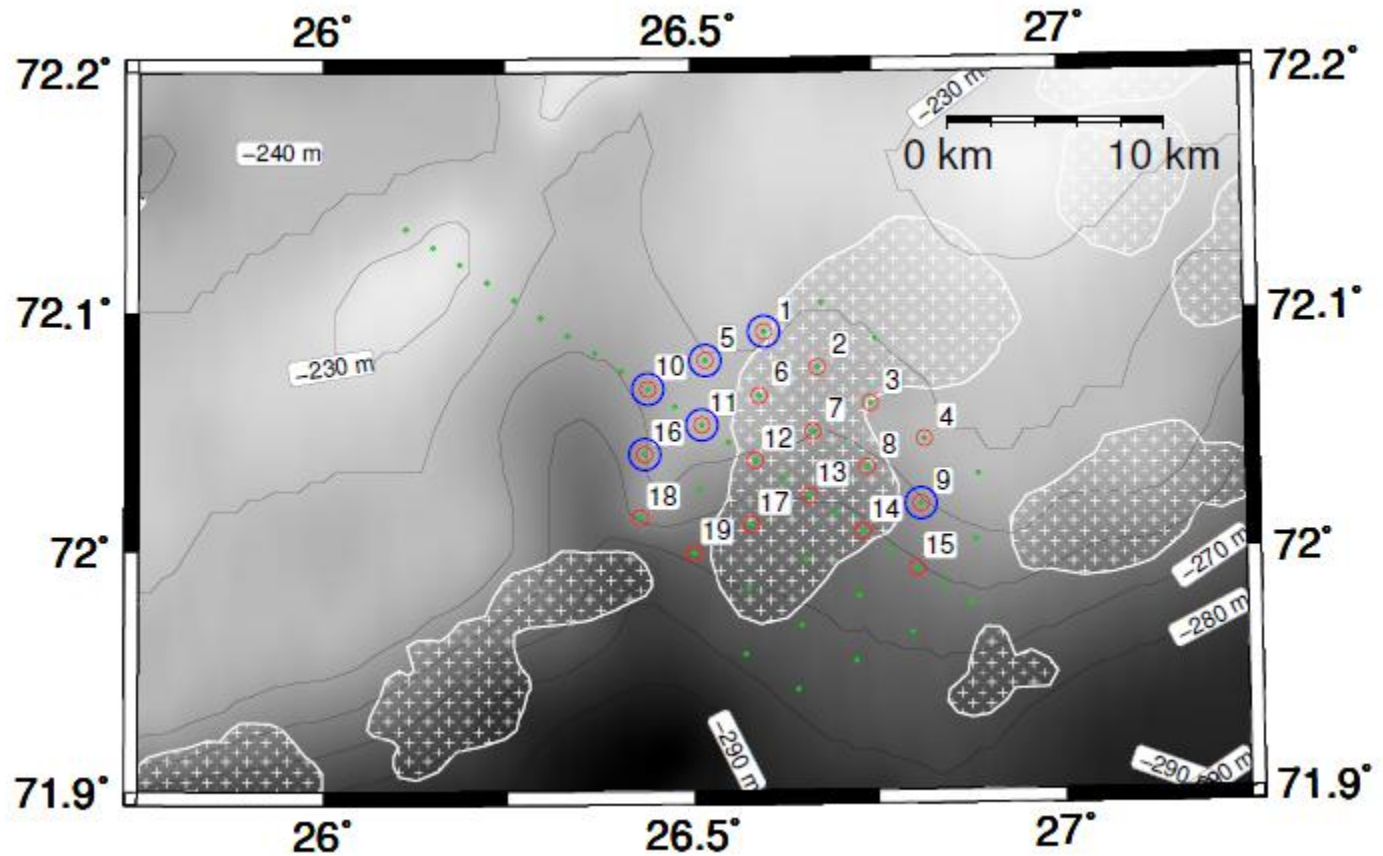


Salt basin:

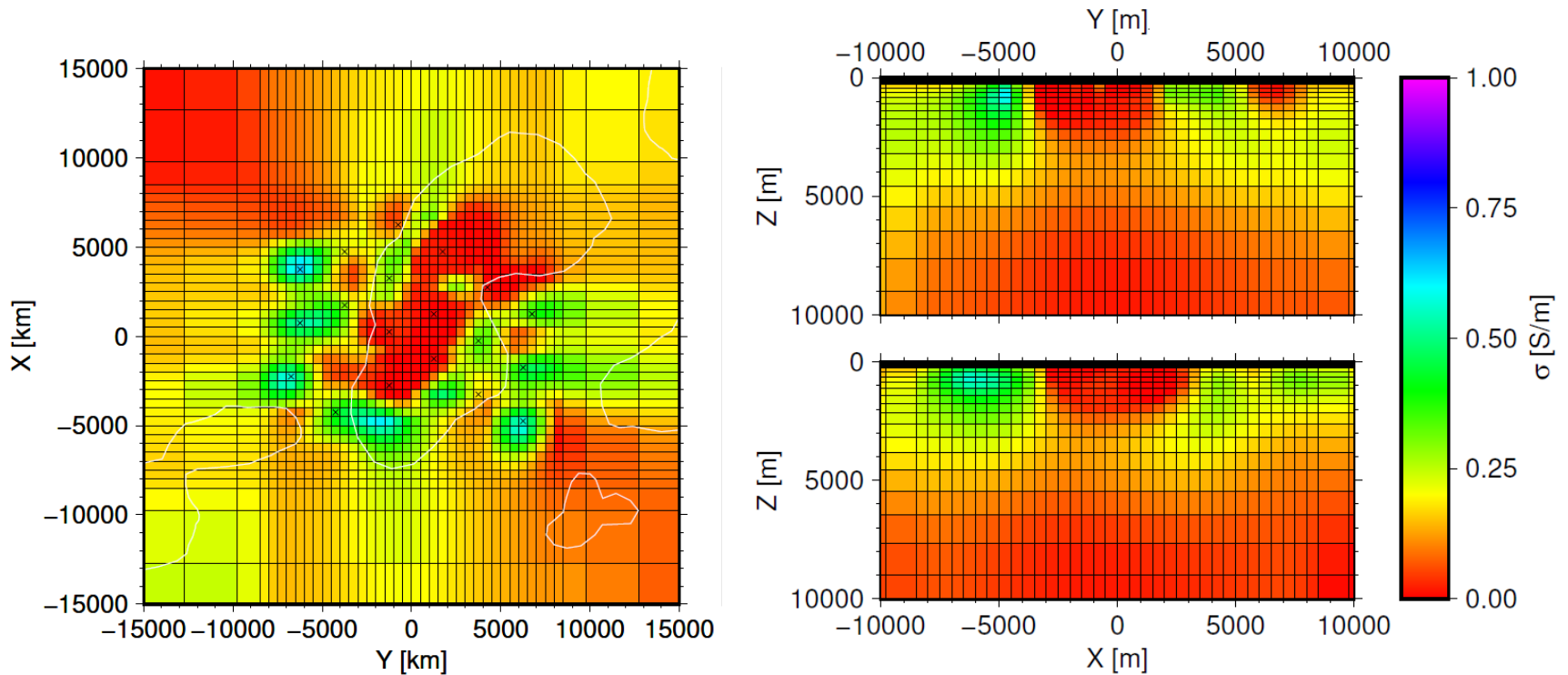
- ~30 salt bodies
- Salt walls in North
- Diapirs in South
- Two wells



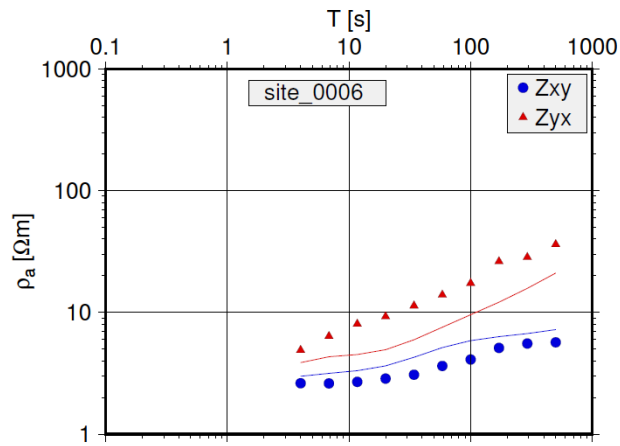
Real data example: Jupiter



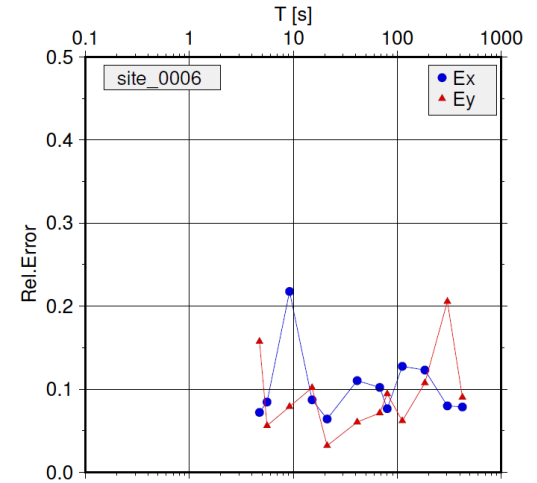
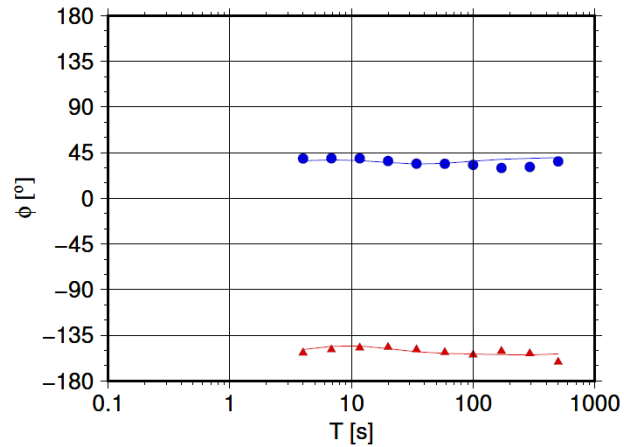
Field inversion



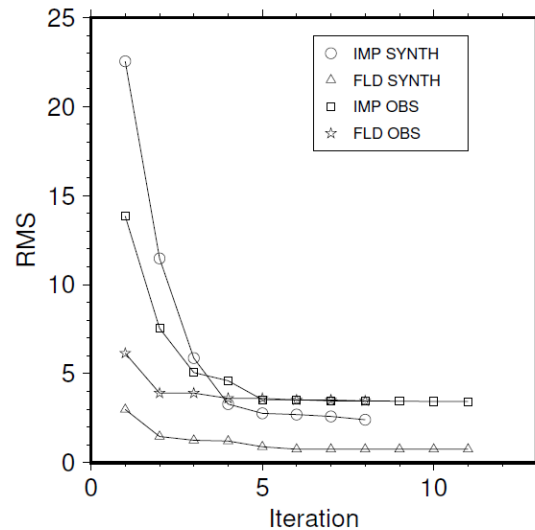
Data fit and error



Impedance inversion



Field inversion



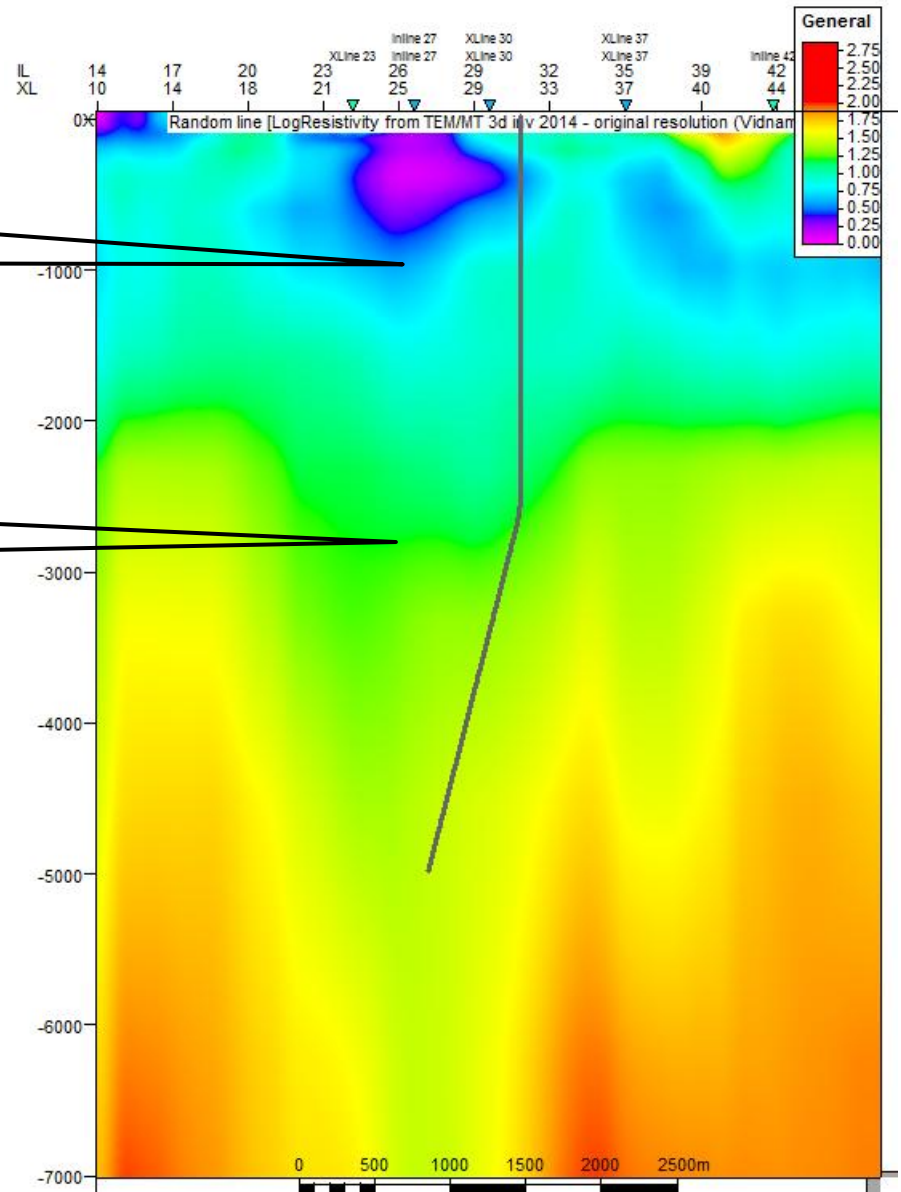
Reykjanes:

3D MT inversion

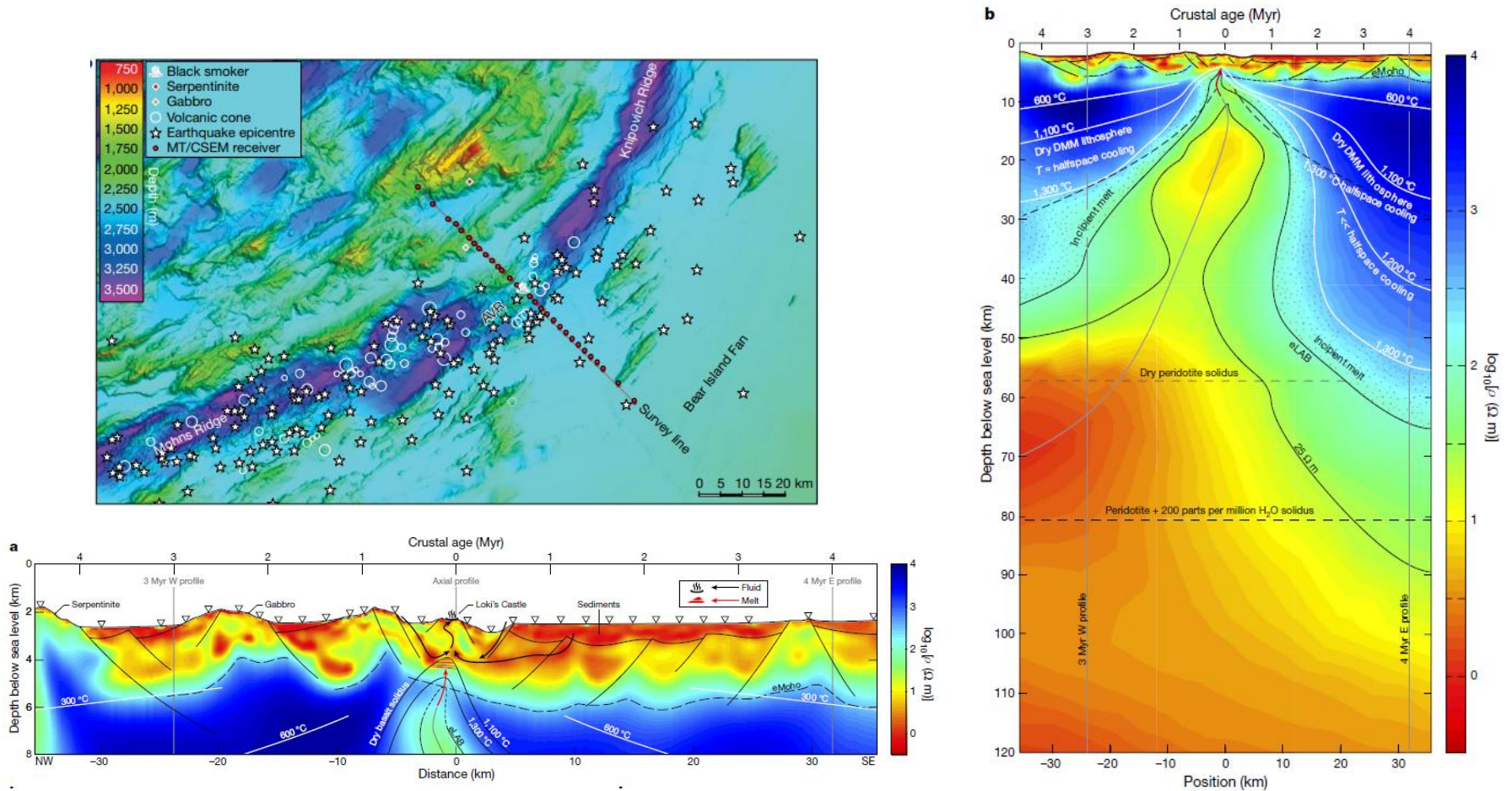
Low resistivity => Low temperature
(smectite diagenesis)

Low resistivity => High temperature

- 3D MT + TEM inversion by ISOR (Ragna Karlsdóttir)
- MT inversion code by Siripunvaraporn (2005); minimum norm method



Mid-Atlantic Ridge CSEM and MT inversion



Figures from Johansen et al., Nature, 2019